

ERAP1 × axial spondyloarthritis

AXIS TARGET DECISION BRIEF | Executive decision | v1.0

Assessment: 5 October 2026. Frozen evidence snapshot through 5 October 2026; selected corpus, not a comprehensive literature cutoff. Primary disease evidence concerns ankylosing spondylitis (AS); broader axSpA applicability is unassessed.

CURRENT AXIS POSITION

Scientific position: **PARTIALLY SUPPORTED** for investigating context-specific ERAP1 modulation. Therapeutic benefit from ERAP1 inhibition is not established in the current AXIS evidence scope.

Strongest evidence: AS genetic association and HLA-context interaction, plus source-reported peptide-processing and cellular perturbation effects.

Strongest concern: cellular phenotypes lack established direct engagement; phenotype direction differs between cell systems.

Critical uncertainty: Do the phenotype-producing compounds directly engage ERAP1 in cells at those exposures?

Recommended next experiment: Matched-genotype chemical-genetic experiment with an engagement readout.

Current decision: **RESOLVE CRITICAL UNCERTAINTY FIRST.**

Why a scientific decision-maker should care

ERAP1 has a plausible evidence chain to AS biology, but attributing a compound-induced phenotype to cellular ERAP1 is the unresolved step that determines whether to invest in that chemical strategy.

AXIS INFERENCE: bounded decision framing from frozen evidence. Recommendation wording is an AI suggestion grounded by deterministic rules; independent scientific acceptance remains pending.

Evidence chain & chemical opportunity



*Experimental structure supported: therapeutic relevance and druggability unestablished.

Evidence layer	AXIS position	Key evidence	Main limitation	Decision relevance
Genetics	SUPPORTED	AS association; HLA interaction	Association ≠ therapeutic direction	Biological rationale
Target-disease	PARTIAL	Genetics + cellular mechanism	Broader axSpA unassessed	Scope program
Mechanism	SUPPORTED	Peptide trimming / repertoire	Allotype and substrate context	Avoid global activity rule
Perturbation	MIXED	Genetic + chemical effects	Opposite FHC directions	Test attribution
Structure	SUPPORTED*	3QNF open state	Missing coordinates; no drug ligand	Bound structure use
Chemical matter	PARTIAL	Maben 1-3	Small set; identity caveats	Reference chemistry
Biochemical	SUPPORTED	Context-specific activity	IC50, AC50, Ki differ	Fix assay context
Cellular	PARTIAL	FHC / presentation / Th17	Systems are not equivalent	Separate phenotype
Engagement	INSUFFICIENT	Not assessed directly	Decision-blocking gap	Measure in cells
Selectivity	INSUFFICIENT	Counter-screen bounds	Comparability unresolved	No ratio claim
Translation	INSUFFICIENT	AS T-cell coculture	No established disease rescue	Further validation
Clinical	NOT ASSESSED	No efficacy indexed	Selected corpus only	No efficacy conclusion

AXIS editorial layer states are categorical summaries, not numeric maturity scores. *Structure existence does not establish druggability. FHC = free heavy chain.

CHEMISTRY: three experimental Maben compounds; three computational exploration candidates (bestatin, captopril, vorinostat). Candidate identity is externally verified, but ERAP1 pharmacology is unestablished. Docking was not executed.

MODEL NOT BUILT: 15 measurements, seven assays, nine distinct assay/endpoint datasets; all nine refused predictive modelling. Identity, prediction and experiment stay separate.

The next decision

THE DECISION BOTTLENECK

Biochemical activity + cellular phenotype



On-target / off-target / indirect / context-dependent explanations



Direct cellular engagement remains unestablished



Matched-genotype engagement + phenotype experiment



Strengthen, weaken, revise mechanism, or repeat after technical validation

WHY THIS EXPERIMENT NEXT?

- It separates three pairs of viable explanations using engagement and genetic dependency together.
- It can weaken the preferred on-target explanation, rather than merely gather supporting observations.
- It connects distinct interpretable outcomes to different scientific actions.
- It uses a target-proximal endpoint with genetic and viability controls; engagement alone would not establish phenotype dependency.

Outcome	Interpretation and decision consequence
A: engagement + dependency + rescue	Strengthens ERAP1-dependent action for this compound and context. Selectivity, indirect mechanism and broader translation remain open.
B: phenotype persists without ERAP1	Weakens ERAP1 attribution for this compound; pursue another chemotype or explanation. Not universal target failure.
C: invalid assay/model or toxicity	Non-interpretable. Keep uncertainty open; validate controls and repeat.

WHAT WOULD CHANGE OUR MIND?

Strengthen: engagement aligned with active exposure, genetic dependency and rescue. Weaken: valid engagement-negative result or ERAP1-independent phenotype. Invalidate the local dependency interpretation: reproducible phenotype persistence in a validated ERAP1-null model. Leave unresolved: failed controls, toxicity, or positive engagement without dependency.

Prerequisites: validated chemical identity, engagement assay and null/rescue model; a reference engaged ligand and inactive analogue are not established in this corpus. Cost, timing and local feasibility are unknown.

Selected references: Evans 2011 (doi:10.1038/ng.873); Chen 2014 (doi:10.1002/art.38249); Chen 2016 (doi:10.1136/annrheumdis-2014-206996); Tran 2016 (doi:10.1016/j.molimm.2016.04.002); Maben (doi:10.1021/acs.jmedchem.9b00293). Independent scientific review pending.

Genetics & mechanistic scope

SOURCE EVIDENCE | Genetic association

ERAP1 polymorphisms were associated with ankylosing spondylitis risk in HLA-B27-positive individuals in this study. [PMID:21743469]

ERAP1 rs30187 was also associated with ankylosing spondylitis in HLA-B*40:01 carriers without HLA-B*27. [PMID:25994336]

SOURCE EVIDENCE | Peptide processing

Purified ERAP1 trimmed a model 13-residue precursor to a 9-residue product and then trimming stopped in that assay. [PMID:16286653]

ERAP1 processing activity depended on the substrate C-terminal residue in purified-enzyme assays. [PMID:16286653]

ERAP1 silencing decreased the proportion of 9-residue HLA-B27-bound peptides in HeLa.B27 and C1R.B27 cells. [PMID:24504800]

ERAP1 silencing increased the proportion of longer 11–13-residue HLA-B27-bound peptides in the tested cell lines. [PMID:24504800]

The K528R ERAP1 variant reduced KK10-specific CTL recognition relative to wild type after extended KK10 minigene expression. [PMID:24504800]

AXIS INFERENCE | Therapeutic hypothesis

Pharmacological modulation of ERAP1 can alter HLA-B27-related antigen-processing biology in a direction relevant to axial spondyloarthritis in appropriate genetic and cellular contexts.

The formulation is an AI suggestion pending researcher approval. Genetic association and peptide-processing effects justify investigation but do not establish the therapeutic direction, a favorable patient phenotype, or benefit from reducing total activity.

PROGRAM BOUNDARY

AS is the disease population indexed in the association studies. ERAP1 genotype, HLA context and substrate dependence constrain extrapolation. An allele-dependent presentation result is not evidence for an allele-specific drug.

Perturbation & competing observations

Perturbation	Source observation	Boundary
Genetic silencing; HeLa.B27 / C1R.B27	Reduced surface FHC; altered HLA-bound peptides	Genetic depletion is not partial pharmacological inhibition
DG013A; HeLa.B27 / C1R.B27	Reduced surface FHC	DG013A inhibits other aminopeptidases; substance identity unresolved
APC / AS CD4+ coculture	Suppressed Th17 expansion after silencing or inhibition	Ex vivo response; no in vivo efficacy established
shRNA; U937.B27	Increased HC10-reactive FHC and disulfide-linked dimers	Different cell background, HLA repertoire and assay
Maben 2 / 3; engineered HeLa	Reduced antigen presentation	Model mouse MHC-I context, not HLA-B27 engagement

Where the evidence disagrees

Chen reports reduced surface FHC after ERAP1 silencing in HeLa.B27/C1R.B27; Tran reports increased surface FHC in U937.B27. The Decision Engine retains the directional disagreement. It does not declare either study false or pool their effects.

U937.B27 carries transfected HLA-B*27:05 alongside endogenous HLA-B*18:01 and HLA-B*51:01; the frozen curation records ERAP1 rs30187/rs27044 heterozygosity. These contextual differences are possible explanations, not proven causes of the discrepancy.

Intervention strategies

Catalytic reduction is a testable hypothesis, not an established treatment. Indexed Maben activity supports substrate-dependent biochemical modulation. Allosteric/conformational explanations remain mechanistic interpretations requiring binding evidence. Allotype-dependent modulation is a question motivated by context. Expression/degradation as a therapeutic strategy is not established in the current AXIS evidence scope; silencing is a perturbation observation.

Experimental structural evidence

SOURCE EVIDENCE | PDB 3QNF

Human ERAP1 open-state crystal structure. X-ray diffraction, 3.0 Å resolution; deposited 2011-02-08, released 2011-02-23. Frozen provider revision 2.2 (2024-11-20).

Construct and canonical mapping

Deposited engineered entity 1, expressed in *Trichoplusia ni*. AXIS maps canonical UniProt Q9NZ08 positions 1-941 by provider-segment projection and sequence verification. The stored construct has 13 inserted expression-tag positions and no recorded substitutions or deletions. Isoform relation is a sequence-checksum inference, not a provider isoform claim.

Chain	Canonical mapped	Coordinates	Unresolved mapped	Inserted positions
A	941	696	245	13
B	941	802	139	13
C	941	800	141	13

Canonical mapping is not coordinate completeness. Missing coordinates do not establish physical truncation. Chain A is the computational campaign input; B and C are retained as distinct structural instances.

Zinc and computational site

The frozen coordinate package records zinc ions, glycans and waters. Campaign preparation retains zinc and records removal of NAG and water. The campaign defines an 8 Å shell around zinc in chain A as a computational site hypothesis. No drug-like ligand is bound in this campaign reference.

AXIS INFERENCE | Limits

The open state is not established as the disease-relevant conformation. This structure does not establish a validated therapeutic pocket, druggability, occupancy, biochemical potency or disease efficacy. Metal proximity is not evidence of chelation; standard docking would not resolve zinc coordination.

Experimental chemistry: measurements

SOURCE EVIDENCE | Maben primary publication, doi:[10.1021/acs.jmedchem.9b00293](https://doi.org/10.1021/acs.jmedchem.9b00293). All 15 indexed measurements are reproduced below without pooling endpoints or censoring boundaries.

Compound	Assay / endpoint	Source value	Interpretation boundary
Maben 1	ERAP1-AMC / IC50	9.2 μ M	Context-specific; not engagement
Maben 1	ERAP1-pNA / Ki	51.7 μ M	Context-specific; not engagement
Maben 2	ERAP1-AMC / IC50	6.9 μ M	Context-specific; not engagement
Maben 2	ERAP2-AMC / IC50	>200 μ M	Censored bound, not exact 200
Maben 2	LNPEP-AMC / IC50	>200 μ M	Censored bound, not exact 200
Maben 2	ERAP1-WK10 / IC50	6.9 μ M	Partial inhibition; <90% maximum
Maben 2	ERAP1-pNA / Ki	12.8 μ M	Context-specific; not engagement
Maben 2	HeLa-presentation / IC50	45 μ M	Presentation phenotype, not binding
Maben 3	ERAP1-AMC / AC50	4.1 μ M	Activation, not inhibition IC50
Maben 3	ERAP1-AMC / fold activity	4.1 fold	Fold activity, not potency
Maben 3	ERAP2-AMC / IC50	>200 μ M	Censored bound, not exact 200
Maben 3	LNPEP-AMC / IC50	>200 μ M	Censored bound, not exact 200
Maben 3	ERAP1-WK10 / IC50	5.3 μ M	Context-specific; not engagement
Maben 3	ERAP1-LF9 / Ki	3.8 μ M	Context-specific; not engagement
Maben 3	HeLa-presentation / IC50	1 μ M	Presentation phenotype, not binding

AMC: ERAP1 L-AMC; ERAP2 R-AMC; mouse LNPEP/IRAP L-AMC. WK10, LF9 and pNA are distinct substrates. Ki uncertainty: compound 1 pNA $\pm$ 4.3 μ M; compound 2 pNA $\pm$ 1.0 μ M; compound 3 LF9 $\pm$ 0.7 μ M; uncertainty type unspecified.

Chemical identity & selectivity

SOURCE EVIDENCE | Identity conflicts preserved

Maben compound 2 is curated as the source urea C₂₁H₃₀N₄O₃ from Scheme 2 and synthesis/HRMS. ChEMBL CHEMBL4456470 represents a guanidine C₂₁H₃₁N₅O₂ and was rejected as a compound mapping. Manual structure transcription still requires independent chemical review.

Provider ≤ 200 μ M off-target records conflict with primary >200 μ M entries. AXIS retains the provider record and uses the primary censored values. The provider IL1 receptor-antagonist mapping was rejected because it is not mouse IRAP Q8C129.

Compound 1 stereochemistry is unresolved in the campaign input; tested substance forms and assay constructs are not fully resolved. DG013A remains a reported perturbagen label and has not been linked to one validated CompoundIdentity. Its cellular observations cannot be attached to Maben biochemical measurements.

AXIS OBSERVATION | Selectivity

Compounds 2 and 3 have ERAP2 and mouse IRAP IC₅₀ bounds >200 μ M. ERAP1 assays differ in substrate, endpoint, construct or conditions. Existing SelectivityAssessments preserve non-comparability and do not establish a quantitative selectivity ratio.

State	What is represented
Measured	Source counter-screen bounds for compounds 2 and 3
Not directly comparable	ERAP1 versus off-target measurements in differing contexts
Not assessed here	Compound 1 numeric off-target bounds; omitted due source/provider inconsistency
Unresolved	Broad off-target profile and selectivity at phenotype-active exposure

DG013A inhibition of ERAP2 and LNPEP is an author-reported caveat in the indexed Chen study, not a new AXIS measurement. No selective therapeutic probe is established.

Cellular pharmacology & engagement

SOURCE EVIDENCE | Separate the layers

Layer	What the indexed corpus supports
Biochemical modulation	Purified-enzyme, substrate-dependent measurements
Cellular phenotype	Surface FHC, peptide repertoire, presentation and coculture Th17 observations
Direct engagement	Not established for phenotype-producing compounds
Target dependency	Genetic observations exist; chemical phenotype attribution remains unresolved
Disease-relevant phenotype	AS T-cell coculture is informative; broader tissue/patient disease rescue is unestablished

DIRECT CELLULAR ERAP1 ENGAGEMENT

NOT ESTABLISHED in the current AXIS evidence scope for the phenotype-producing compounds. Cellular phenotype and biochemical activity do not close this link.

Why this gap controls the decision

Two of the three indexed Maben compounds have biochemical activity and cellular phenotypes; the evidence remains separated by compound. Direct cellular engagement has not been assessed. Selectivity is unresolved, leaving an off-target explanation viable. DG013A also has unresolved substance identity.

The Decision Engine therefore selects target engagement as decision-blocking and directly testable. It retains target dependency, genetic context, reproducibility, mechanistic bridge and translation as additional uncertainties. More literature or more unlinked biochemical potency measurements would not establish cellular attribution.

AXIS INFERENCE | The recommendation is exploratory and pending independent acceptance. No engagement experiment has been performed by AXIS; synthetic result-loop demonstrations are excluded from this brief.

Competing explanations

AXIS INFERENCE | Exact current explanation labels and states: on-target is partially supported; the other three are viable. All remain under consideration; none is an accepted experimental result.

Explanation / current state	Supporting or compatible evidence	Evidence against / missing evidence
Context-dependent effect / VIABLE	Directional FHC disagreement and unmatched contexts	Allotype/subtype cause unproven; matched comparison needed
Indirect pathway / VIABLE	Proximal functional modulation insufficiently shown	No direct pathway discrimination; phenotype alone insufficient
Off-target activity / VIABLE	DG013A counter-target caveat; selectivity unresolved	No causal off-target mechanism established; genetic effects do not exclude it
On-target ERAP1 modulation / PARTIALLY_SUPPORTED	Biochemical activity and chemical phenotype are compatible	Missing direct engagement and chemical-genetic dependency

Source observations versus interpretation

Biochemical activity is observed in the source assays; attribution of the cellular phenotype to on-target action is an inference to be tested. An unresolved off-target profile is not proof of off-target causation. Directional disagreement across cell systems motivates a context-dependent explanation without establishing its cause.

What the recommended experiment separates

Engagement plus an ERAP1-null/rescue comparison can distinguish on-target, off-target and indirect explanations across the stored outcome scenarios. An engagement-positive, dependency-positive result still does not distinguish every ERAP1-dependent downstream route. A single matched background does not establish generality across allotypes or HLA-B27 subtypes.

The underlying state preserves supporting, unresolved and context-limiting links by evidence ID and rule ID. The appendix and state snapshot expose those links so that a reviewer can disagree with the inference without losing the source chain.

Computational discovery: bounded panel

RESEARCHER HYPOTHESIS | A molecule with a metal-binding group that reaches the catalytic zinc region may inhibit catalytic trimming. This chemical hypothesis does not assume that inhibition treats disease.

Candidate / role	Computational evidence	Reason to test / reason against	Required validation
Captopril / exploration	Distinct cluster; reference similarity 0.125 (Morgan/Tanimoto). No docking evidence.	Test zinc-region chemical hypothesis; no indexed ERAP1 activity, engagement or selectivity.	Identity/form check; fixed biochemical context; concentration response and interference counterscreen; then engagement.
Bestatin / exploration	Distinct cluster; reference similarity 0.1449 (Morgan/Tanimoto). No docking evidence.	Test zinc-region chemical hypothesis; no indexed ERAP1 activity, engagement or selectivity.	Identity/form check; fixed biochemical context; concentration response and interference counterscreen; then engagement.
Vorinostat / exploration	Distinct cluster; reference similarity 0.1724 (Morgan/Tanimoto). No docking evidence.	Test zinc-region chemical hypothesis; no indexed ERAP1 activity, engagement or selectivity.	Identity/form check; fixed biochemical context; concentration response and interference counterscreen; then engagement.

AXIS OBSERVATION | PubChem identity verification: bestatin CID 72172; captopril CID 44093; vorinostat CID 5311. Identity verification is not ERAP1 pharmacology evidence.

Panel interpretation

No computational candidate reached the exploitation similarity threshold; the exploitation role is empty, not filled.

The panel is chemically diverse but tests 1 hypothesis; chemical diversity is not hypothesis diversity.

Hypothesis hypothesis:erap1:maben-series-analog is not tested by any panel member of this chemical space.

No experimentally supported inactive control is indexed; none was fabricated.

Docking status

METHOD UNAVAILABLE; VALIDATION NOT ESTABLISHED; EXECUTION NOT ATTEMPTED; NO RESULT. The replay found no docking engine or PDBQT preparation toolchain. The frozen campaign has no validated target/site docking method, and standard scoring does not represent metal coordination.

The candidates are computationally prioritized exploration molecules, not active compounds, validated binders or leads. Similarity values are chemical metrics, not target scores, probability estimates or potency predictions.

Chemical learning: what can be learned?

CHEMISTRY → NEXT TEST

Experimental reference chemistry (Maben 1-3)

+ computational exploration panel (bestatin, captopril, vorinostat)

+ model-readiness refusal

↓

Controlled biochemical testing for new candidates; engagement/dependency remains the main program decision.

PREDICTIVE MODEL STATUS | NOT BUILT

All nine frozen ERAP1 chemistry datasets are not eligible. The indexed chemistry is too small and fragmented across assay contexts for defensible predictive modelling under the current AXIS policy.

Dimension	Current AXIS observation
Compounds / measurements	3 experimental compounds / 15 source measurements
Assays / contexts	7 assays across 3 targets; 9 endpoint-context datasets
Comparability	36 dataset-pair assessments; none directly comparable
Observed SAR	No matched molecular pairs; no shared-core pair in this set
Scaffold records	15 algorithmic scaffold-group records, not 15 experimental SAR relationships
Inferred SAR	No SAR hypothesis proposed
Readiness	6 SAR_ONLY; 3 INSUFFICIENT_DATA; all model-ineligible

What the chemistry can teach the program

The indexed substrate-dependent activation/inhibition and cellular presentation results establish the need to keep endpoints separate. They do not establish a transferable potency ranking or a predictive structure-activity relationship.

The most populated ERAP1 L-AMC IC50 context contains two exact compounds on two scaffolds, spanning about 0.125 log10 units. A useful expansion would measure more compounds in one fixed context, diversify scaffolds and activity range, include actual matched pairs and make off-target measurements comparable.

AXIS refuses to manufacture a QSAR result when its indexed evidence does not support one. These limits describe this dataset and modelling policy; they are not universal biological laws.

Recommended experiment & outcome tree

AI SUGGESTION | Deterministically selected

Matched-genotype chemical-genetic experiment with an engagement readout

Scientific question: Do the phenotype-producing compounds directly engage ERAP1 in cells at those exposures?

Context: HLA-B27-positive cell system with wild-type, ERAP1-depleted or null, and ERAP1-rescued arms in a matched genetic background

Intervention: Compound at phenotype-active exposures versus vehicle and an inactive analogue in each arm; ERAP1 genetic depletion and re-expression as the genetic comparison.

Primary endpoint: Compound-dependent ERAP1 engagement at the exposures that produce the phenotype, together with phenotype attenuation in ERAP1-null cells

Controls: vehicle; matched ERAP1-null/depleted cells; inactive analogue if established; validated engaged reference ligand required. Viability measured separately; ERAP1 re-expression tests restoration.

Outcome	Interpretation / decision
S1 / supportive	Engagement is detected at phenotype-active exposures in wild-type cells; the phenotype is attenuated in ERAP1-null cells and restored by ERAP1 re-expression. If engagement and ERAP1-dependence are both observed, the current evidence would support continuing the ERAP1 strategy for this compound in this context, with selectivity and context still to be addressed.
S2 / negative	Engagement is detected in wild-type cells, but the phenotype persists unchanged in ERAP1-null cells. If the phenotype persists without ERAP1, the compound would no longer support an ERAP1-dependent strategy and alternative chemotypes would need their own engagement and dependency evidence.
S3 / alternative	No engagement signal is detected in a validated assay, yet the phenotype is attenuated in ERAP1-null cells. If the phenotype is ERAP1-dependent without detectable engagement, the mechanistic model would change toward an indirect route and the strategy would need re-justification.
S4 / alternative	Engagement and ERAP1-dependence differ between HLA-B27 backgrounds or ERAP1 allotypes. If engagement or dependence varies with background, the next evidence needed would be a matched allotype and HLA-B27 subtype comparison before any strategy decision.
S5 / non_interpretable	The null or rescue model is not validated, or toxicity confounds the phenotype. If the experiment is non-interpretable, the uncertainty would remain open and the model and controls would need to be validated before repeating it.

Prerequisites remain unestablished: compound identity where unresolved, engagement readout and validated null/rescue model. Local cost and time are not estimated. One molecular cellular system does not test disease rescue or resolve full selectivity.

Falsification, decision & validation limits

WHAT WOULD CHANGE OUR MIND?

- Strengthen: cellular engagement at active exposure plus loss of phenotype in null cells and rescue after re-expression.
- Weaken: phenotype persists without ERAP1; or no engagement in a validated assay at phenotype-active exposures.
- Invalidate the specific ERAP1-dependent attribution: a reproducible ERAP1-independent phenotype in a valid matched model. This does not invalidate every ERAP1 intervention.
- Revise mechanism: dependency without detectable engagement favors an indirect route under the stored scenario.
- Leave unresolved: failed assay/model controls, toxicity, or engagement without dependency.

CURRENT DECISION | RESOLVE CRITICAL UNCERTAINTY FIRST

WHY: cellular attribution is unestablished while multiple explanations remain viable.

NEXT: establish the prerequisites and perform the matched-genotype engagement/phenotype experiment.

DO NOT CONCLUDE YET: inhibition treats axSpA; computational candidates are active; an ERAP1 QSAR is valid; the phenotype predicts clinical efficacy.

Retrospective validation

The replayed ERAP1 T2011, T2014 and T2016 cases all conclude: future evidence did not test the decision. Later publications informed other uncertainties but did not resolve the engagement question. AXIS did not force hindsight confirmation. These development benchmarks do not establish prospective predictive validity.

Scientific limitations

Small purposefully selected corpus; not a systematic review. AS evidence cannot automatically extend to all axSpA. HLA, allotype, cell and substrate contexts differ. Engagement, chemical dependency, selectivity and therapeutic direction remain unresolved. Chemistry is small; no ERAP1 QSAR or prospective validation of candidates. Clinical efficacy and evidence that ERAP1 inhibition treats axSpA are not established in the current AXIS scope. Independent scientific and medicinal-chemistry review is pending.

Primary references & evidence drill-down

Source	Frozen primary reference	DOI
PMID:21743469	Interaction between ERAP1 and HLA-B27 in ankylosing spondylitis implicates peptide handling in the mechanism for HLA-B27 in disease susceptibility.	10.1038/ng.873
PMID:25994336	Major histocompatibility complex associations of ankylosing spondylitis are complex and involve further epistasis with ERAP1.	10.1038/ncomms8146
PMID:16286653	The ER aminopeptidase, ERAP1, trims precursors to lengths of MHC class I peptides by a "molecular ruler" mechanism.	10.1073/pnas.0500721102
PMID:24504800	Critical role of endoplasmic reticulum aminopeptidase 1 in determining the length and sequence of peptides bound and presented by HLA-B27.	10.1002/art.38249
PMID:26130142	Silencing or inhibition of endoplasmic reticulum aminopeptidase 1 (ERAP1) suppresses free heavy chain expression and Th17 responses in ankylosing spondylitis.	10.1136/annrheumdis-2014-206996
PMID:27107845	ERAP1 reduces accumulation of aberrant and disulfide-linked forms of HLA-B27 on the cell surface.	10.1016/j.molimm.2016.04.002
PMID:31841350	Maben et al. Discovery of Selective Inhibitors of Endoplasmic Reticulum Aminopeptidase 1. J Med Chem 2020;63:103-121.	10.1021/acs.jmedchem.9b00293

Auditable conclusion chains

Conclusion	Claim → experiment → source
AS genetic relevance	C01/C02 → association / replication → Evans / Cortes
Peptide-processing context	C03-C07 → trimming / immunopeptidome / variant presentation → Chang / Chen 2014
Phenotype disagreement	C08 vs C12/C13 → HeLa/C1R vs U937 readouts → Chen 2016 / Tran
Biochemical chemistry	measurement:maben:1-15 → substrate-specific assays → Maben, locators in snapshot
Engagement gap	cellular gap IDs → engagement assessments → DecisionState critical uncertainty
Recommended experiment	DecisionState → DECISION-EXP rules → candidate profile and prospective scenarios

Reference validation checks identifier consistency against the frozen indexed packages; it is not a new full-text literature audit. Discovery access varies: abstracts/metadata for several sources; selected full-text methods/results for Chen and Tran; Maben primary PDF factual extract. Original publication rights remain with their owners.

Technical provenance & reproducibility

Assessment date: 2026-10-05. Evidence cutoff: frozen indexed snapshot through 2026-10-05; original source retrieval/curation dates preserved. This date is not a claim that the literature is comprehensively current. AXIS 0.2.1.dev0. Repository: JPais7/AXIS. Template: axis-commercial-brief-1.0.

Source main SHA: 8f4e7797075e2beb25ad80de13d90843ad4e9de2

DecisionState: decision-state:1:cebbae3880aeec90

Decision rules: axis-decision-3

Rule fingerprint: 184a1e2949a61e362545e897de43890014bcdcac7191f621c84fb2b739cecd33

ComputationalCampaign: campaign:erap1:bounded-chemistry-v1@r1

Prioritization fingerprint: 9587589ac4b984f5d502baa83e04634c98ac02e4ff43f51ee69fb2d8b35595e0

ChemicalLearningState: learning-state:AXIS-DD-ERAP1-CURATED-001@r1

Content fingerprint: a098ff64c89db792bb5a77d0b355fce0c095e1dd126efe7fb620917656756b1c

Eligibility policy fingerprint: f0ba60960e3044f2bbb0910242ea3a09f4ced35e0d842ff3a66d1dbc7ed163f1

Frozen package	Version
discovery/erap1-axspa/v1	1.0.0
targets/erap1/uniprot/v1	1.0.0
structures/erap1/3qnf/v1	1.0.0
pharmacology/erap1/v1	erap1-pharmacology-v1
cellular/erap1-axspa/v1	1.0.0
decision/erap1-axspa/v1	1.0.0
benchmarks/retrospective/erap1-axspa/v1	v1 path; no declared semantic version
computational-discovery/erap1/v1	v1 path; no declared semantic version

AXIS modelling policy: minimum 20 exact compounds, 5 scaffolds and 1 log10 activity range; maximum censored fraction 0.5. Other criteria are preserved in state-snapshot.json. These are policy thresholds, not scientific laws.

Reproduce: run extract.py from the source checkout with AXIS dependencies, then build.py with ReportLab and pypdf. The replay creates a temporary fresh Evidence Store with network connections blocked; no synthetic experimental results are imported. Source data, state snapshot, traceability map, package hashes and output hashes are retained in this directory.

AXIS

Drug Discovery Decision Intelligence

From evidence to uncertainty to the next scientific decision.

AXIS is designed to help scientific teams interrogate evidence, expose decision-critical uncertainty and determine what should be tested next. It does not replace experimental validation or scientific judgment.

Know what to test next - and why.

ERAP1 × axial spondyloarthritis | v1.0 | 5 October 2026

READY WITH CONDITIONS for design-partner discussion. AI-assisted curation and proposed experiments remain pending independent scientific review. Scientific approval for external investment diligence has not been obtained.